آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-I Business Mathematics Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part I Business Mathematics. It contains comments on candidates' responses to the 2024 HSSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfill the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Overall, candidates performed well on concepts involving

- Arithmetic series
- Geometric series
- Compound interest
- Simultaneous linear equations.

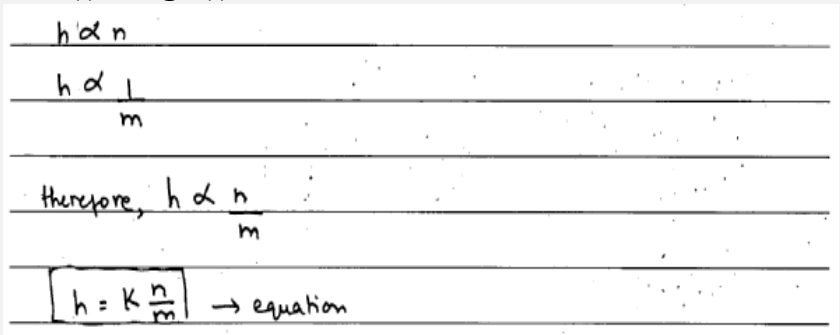
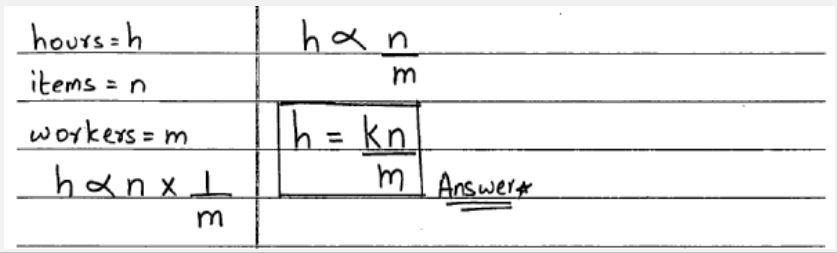
However, there were noticeable struggles with understanding and applying the concept of

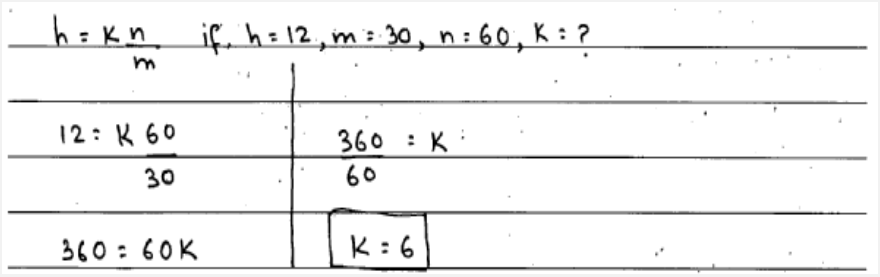
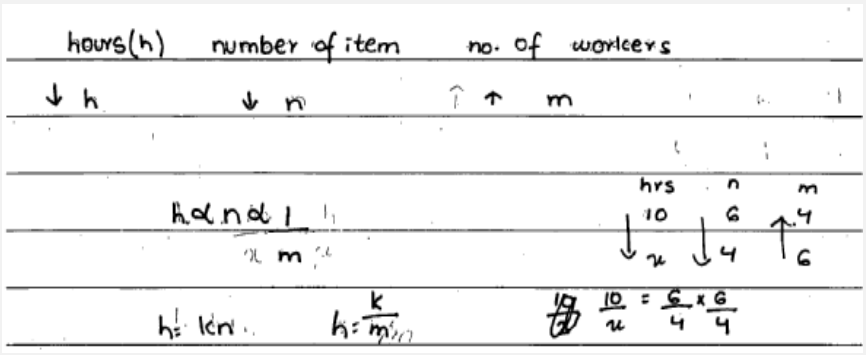
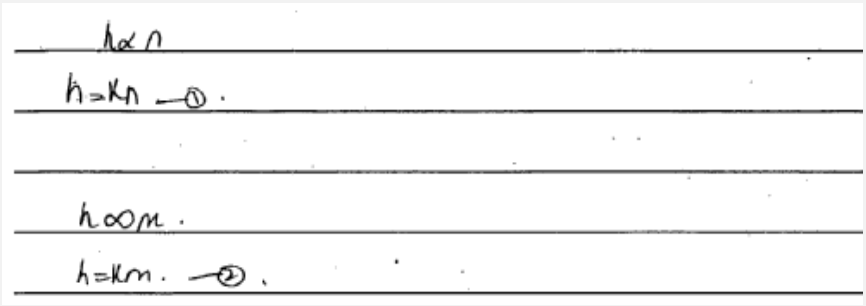
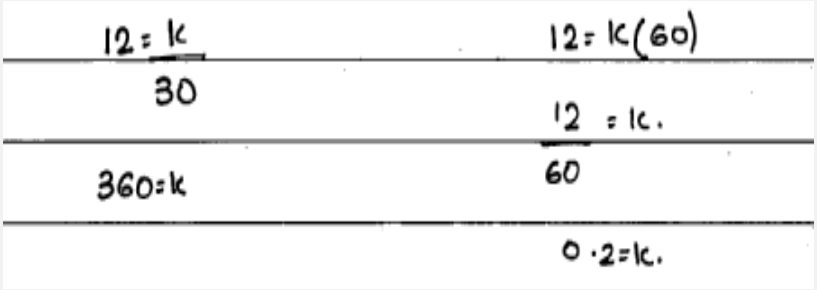
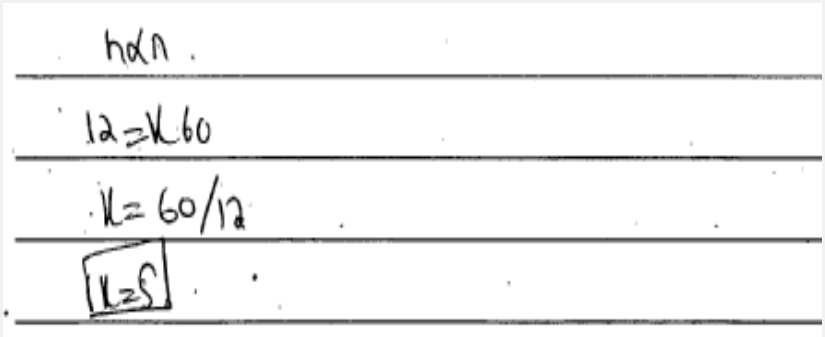
- Proportionality
- Quadratic equations
- Graphing inequalities
- Cramer's rule.

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

Question No. 1	
Question Text	In a factory, the time required in hours (h) is directly proportional to the number of items (n) produced and the time required in hours (h) is inversely proportional to the number of workers (m). i. Using the given information, form an equation for the given variation. ii. It takes 12 hours for 30 men to complete a demand of 60 items of a certain product. Find the value of the constant.
SLO No.	1.2.2
SLO Text	Solve word problems related to direct, inverse and compound (joint and combined) variations.
Max Marks	4
Cognitive Level	A*
Checking Hints	i. 1 mark for making $h \alpha n$ and $h \alpha \frac{1}{m}$ 1 mark for combining the two variations and write as $h \alpha \frac{n}{m}$ and writing in the form of equation $h = k\frac{n}{m}$ ii. 1 mark for substituting the value of h, m and n 1 mark for finding the value of k
Overall Performance	Majority of the candidates demonstrated a good understanding of the concept of proportionality, accurately forming and solving the equation. However, a notable number of candidates struggled with combining the direct and inverse proportional relationships, resulting in incorrect equations and solutions.
Description of Better Responses	In better responses, candidates correctly identified that time (h) is directly proportional to the number of items (n) produced, leading to $h \alpha n$ and inversely proportional to the number of workers (m), leading to $h \alpha \frac{1}{m}$. Combining these, they formed the correct equation $h = k\frac{n}{m}$ and successfully extracted the data from the question. By substituting the values h =12, n = 60 and m =30 into the equation, they determined the value of k correctly as k = 6.
Images of Better Responses	**Part(i) Image (i)**  **Image (ii)** 

	Part (ii) 
Description of Weaker Responses	In weaker responses, candidates struggled to combine the direct and inverse proportional relationships. They either solved the direct and inverse proportions separately or failed to introduce the constant *k*, resulting in incorrect equations. Consequently, they substituted the given values into the wrong equation and were unable to determine the correct value of *k*.
Images of Weaker Responses	**Part (i) Image (i)**  **Image (ii)**  **Part (ii) Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

Teachers may use the following teaching activities to improve the performance in this concept.

- **Step-by-Step Guidance:** Provide step-by-step guidance on how to approach direct proportion problems. Encourage students to clearly define variables and use proportional reasoning.
- **Comparison Exercises:** Create exercises that directly compare direct and indirect proportion scenarios, allowing students to contrast the two concepts and identify their differences.
- **Real-Life Examples:** Illustrate with examples like work hours related to the number of employees and tasks to solidify understanding.
- **Group Activities:** Encourage students to work in groups to solve similar problems, fostering peer learning and collaboration.
- **Visual Aids:** Use graphs and charts to demonstrate proportional relationships and their impacts.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 2a **Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.**	
Question Text	Find the effective rate of interest corresponding to nominal interest rate of 10% per annum compounded i. annually. ii. quarterly.
SLO No.	2.2.4
SLO Text	Calculate effective annual rate of interest.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for the substitution in the formula 1 mark for taking m = 1 ii. 1 mark for the substitution in the formula 1 mark for taking m = 4

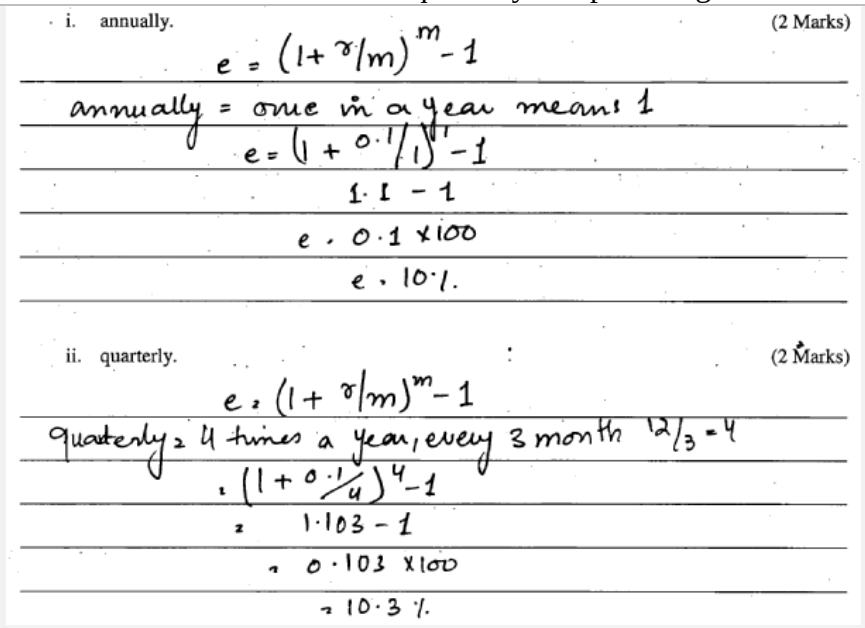
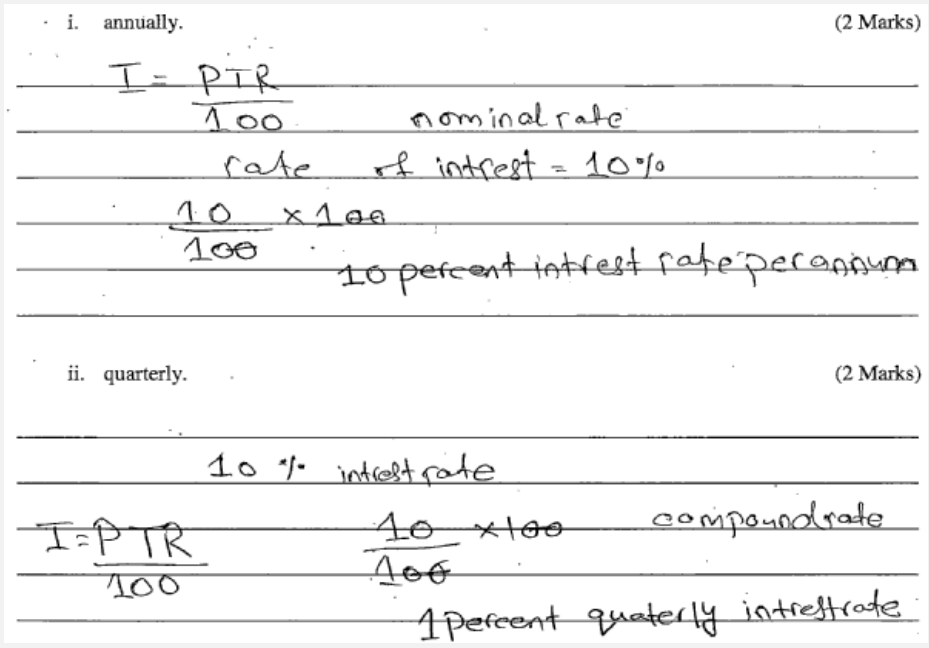
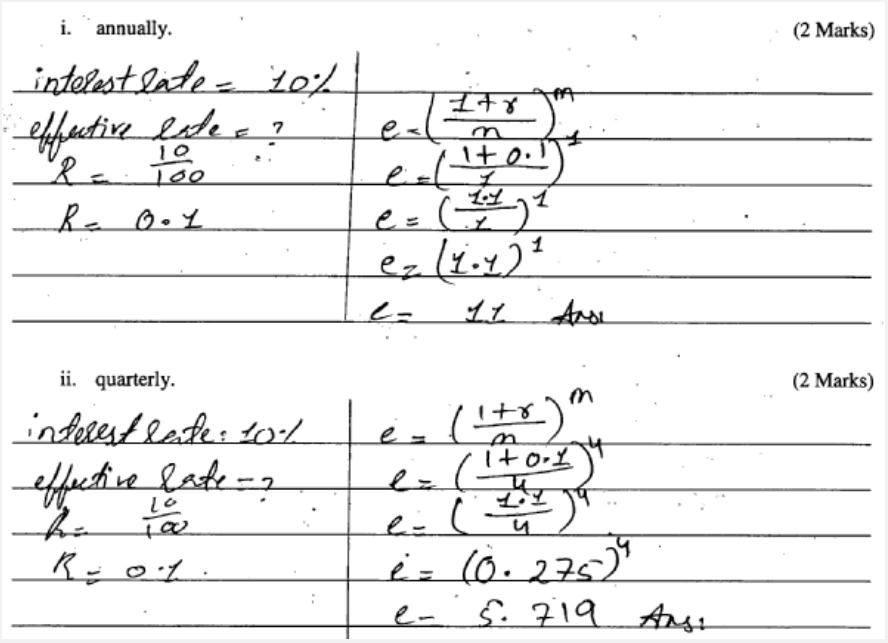
Overall Performance	Many of the candidates attempted this part and a small number of candidates generally performed well in this part as compared to part *b*, with many such responses showing a clear grasp of effective interest rates and their calculations. Weaker responses highlighted confusion between different types of interest rates and compounding periods.
Description of Better Responses	In better responses, the candidates identified the correct formula for the effective annual rate of interest: Annually: $\left(1+\frac{r}{m}\right)^n - 1$ and Quarterly: $\left(1+\frac{0.10}{4}\right)^4 - 1$ They substituted the values correctly, made no calculation errors and understood the difference between annual and quarterly compounding.
Image of Better Response	 i. annually. (2 Marks) $e = (1+r/m)^m - 1$ annually = once in a year means 1 $e = (1+0.1/1)^1 - 1$ $1.1 - 1$ $e = 0.1 \times 100$ $e = 10\%$ ii. quarterly. (2 Marks) $e = (1+r/m)^m - 1$ quarterly = 4 times a year, every 3 month $12/3 = 4$ $= (1+0.1/4)^4 - 1$ $= 1.103 - 1$ $= 0.103 \times 100$ $= 10.3\%$
Description of Weaker Responses	In weaker responses, many candidates confused the effective rate of interest with compound interest, using incorrect formulas such as the simple interest formula. Some applied the correct formula but substituted incorrect values of *m* in part (ii), reflecting a weak understanding of the terms annually and quarterly.
Images of Weaker Responses	**Image (i)**  i. annually. (2 Marks) $I = \frac{PTR}{100}$ nominal rate rate of intrest = 10% $\frac{10}{100} \times 100$ 10 percent intrest rate per annum ii. quarterly. (2 Marks) 10 % intrest rate $I = \frac{PTR}{100}$ $\frac{10}{100} \times 100$ compound rate 1 percent quaterly intrest rate

Image (ii)



i. annually. (2 Marks)

interest rate = 10%
effective rate = ?
$R = \frac{10}{100}$
$R = 0.1$

$e = \left(\frac{1+r}{n}\right)^{m}$
$e = \left(\frac{1+0.1}{1}\right)^{1}$
$e = \left(\frac{1.1}{1}\right)^{1}$
$e = (1.1)^{1}$
$e = 1.1$ Ans

ii. quarterly. (2 Marks)

interest rate: 10%
effective rate = ?
$R = \frac{10}{100}$
$R = 0.1$

$e = \left(\frac{1+r}{n}\right)^{m}$
$e = \left(\frac{1+0.1}{4}\right)^{4}$
$e = \left(\frac{1.1}{4}\right)^{4}$
$e = (0.275)^{4}$
$e = 5.719$ Ans:

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:

Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.

- **Digital Tools:** Use financial calculators or software like Excel for compound interest calculations. Interactive tutorials can provide instant feedback and help clarify misconceptions.
- **Real-Life Examples:** Compare different savings account options with varying compounding periods to demonstrate the practical impact.
- **Class Discussions:** Facilitate discussions on the differences between nominal, effective, and compound interest rates to reinforce concepts.
- **Practice Problems:** Provide a variety of problems in different settings with different compounding periods to build proficiency.

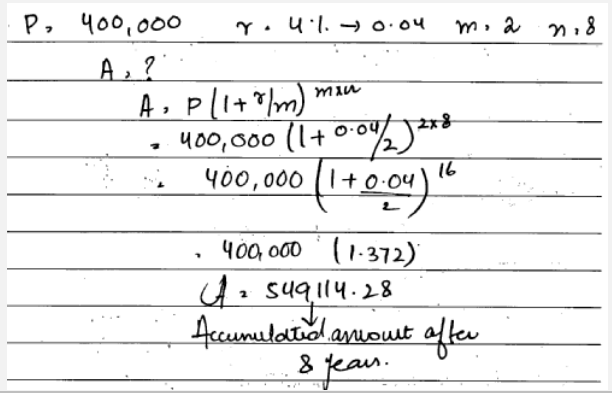
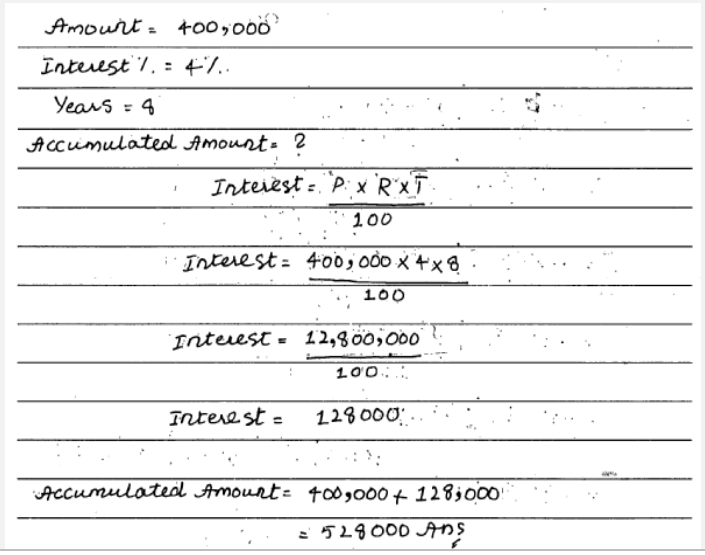
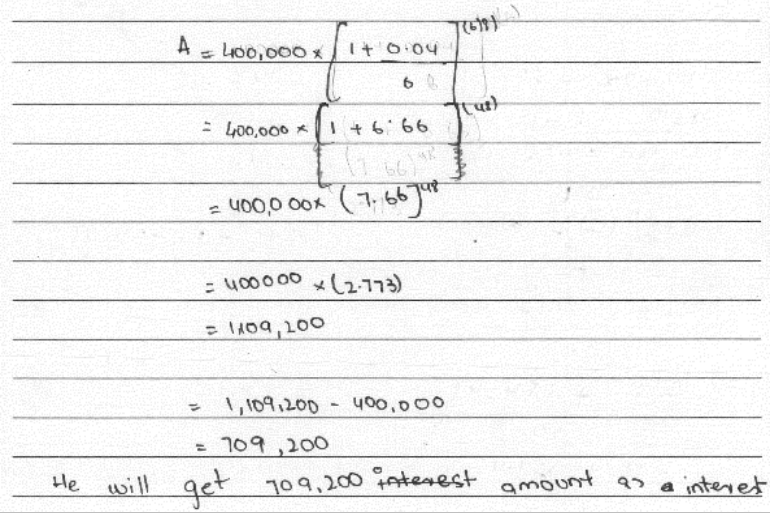
Question No. 2b Candidates were given the choice to attempt any ONE out of the two questions: 2a and 2b.	
Question Text	Saad deposits Rs 400,000 in his savings account at a bank. The account earns 4% interest per year, compounded semi-annually. What will be the accumulated amount in his account after 8 years?
SLO No.	2.2.5
SLO Text	Solve word problems related to the concept of compound interest.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for selecting the correct formula $A = p\left(1+\frac{r}{m}\right)^t$ 1 mark for finding the correct value of $t = 16$ 1 mark for substituting values in the formula 1 mark for calculating the accumulated amount
Overall Performance	This part is well attempted by many of the candidates. Many of the candidates demonstrated a strong grasp of compound interest, correctly applying the relevant formulas. Weaker responses indicate confusion related to compounding periods and formula application.
Description of Better Responses	In better responses, candidates applied the correct formula for compound interest $A = P\,(1+rn)^{nt}$ with $P = 400{,}000$, $r = 0.04$, $n = 2$ (semi-annually) and $t = 8$. They substituted the values accurately, showing a strong understanding of semi-annual compounding and the concept of accumulated amount.
Image of Better Response	
Description of Weaker Responses	In most of the weaker responses, candidates used the simple interest formula instead of the compound interest formula. Others misunderstood the term semi-annual compounding, substituted incorrect values for n or r as 6 and 0.04, and made calculation errors.
Images of Weaker Responses	**Image (i)** 

	Image (ii) 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

By incorporating these interactive teaching strategies, teachers may help students understand both compound and simple interest, apply the concepts effectively, and recognise their importance in various financial scenarios.

- **Digital Tools:** Use online compound interest calculators to practice different scenarios and visualise the growth of investments over time.
- **Real-Life Examples:** Show the growth of an investment over time with different compounding periods using visual aids like graphs and charts.
- **Interactive Simulations:** Implement simulations where students can manipulate variables like interest rate and compounding frequency to see their effects.

Question No. 3	
Question Text	Arham bought x dozen bananas for Rs 600. If he could pay Rs 10 less per dozen, he would have been able to get a dozen more. By converting the above statements into mathematical expressions, show that the quadratic equation is formed as $x^2 + x - 60 = 0$.
SLO No.	4.1.2

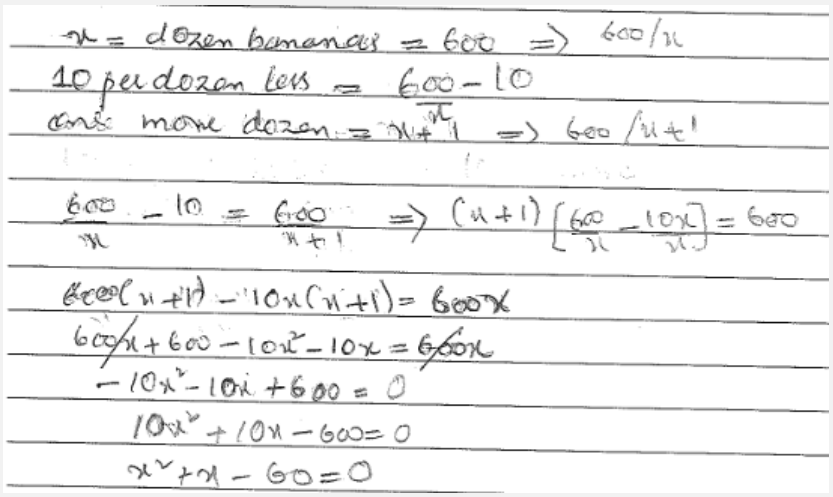
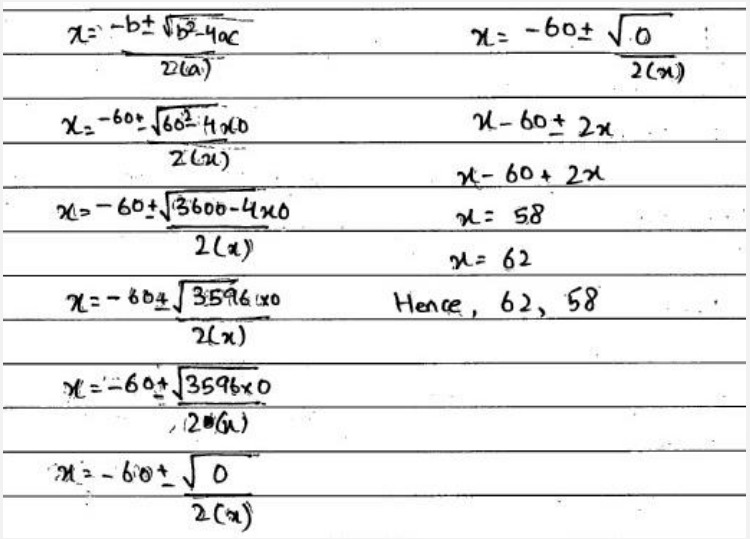
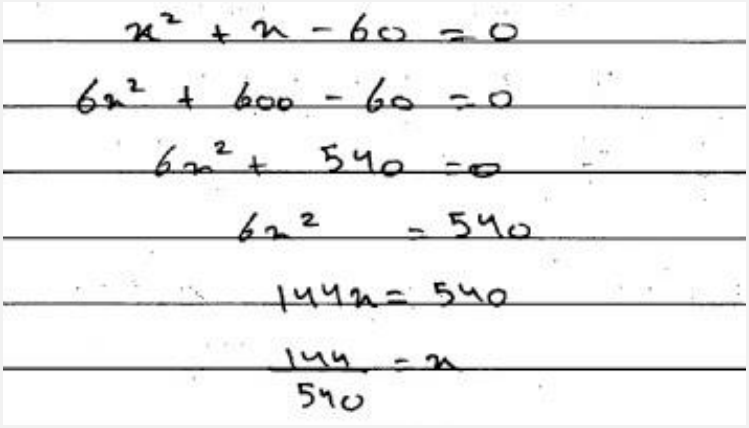
SLO Text	Solve problems based related to quadratic equation.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for writing expression for price per dozen $\frac{600}{x}$ and $\frac{600}{x+1}$ 1 mark for converting the situation into algebraic form as $\frac{600}{x}-10=\frac{600}{x+1}$ 1 mark for writing the expression in standard form as $x^2+x-60=0$
Overall Performance	Few candidates could accurately convert the problem into mathematical expressions and derive the quadratic equation. Many struggled with the initial step, indicating a lack of understanding of translating word problems into equations.
Description of Better Responses	In better responses, candidates correctly identified that $\frac{600}{x+1}$ is the cost of the bananas with an additional dozen. By setting up the equation $\frac{600}{x}-10=\frac{600}{x+1}$ they correctly manipulated the expressions to derive $600(x+1) - 600x = 10x(x + 1)$, leading to the quadratic equation $x^2+x-60=0$.
Image of Better Response	
Description of Weaker Responses	In weaker responses, majority of the candidates struggled to identify the correct expressions for cost per dozen and the new cost with one more dozen. Many did not combine the expressions correctly, resulting in incorrect equations. Some applied the quadratic formula directly to the provided equation without deriving it first.
Images of Weaker Responses	**Image (i)** 

	Image (ii) 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:

Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.

- **Digital Tools:** Use algebra software like Wolfram Alpha to visualise quadratic equations and their solutions.
- **Real-Life Examples:** Create similar problems involving prices and quantities to practice forming and solving equations.
- **Step-by-Step Tutorials:** Provide guided tutorials that break down the process of translating word problems into mathematical expressions.

Question No. 4a **Candidates were given the choice to attempt any ONE out of the two questions: 4a and 4b.**	
Question Text	Solve the given simultaneous equations. $3x^2 + y^2 = 4$ $x - y + 2 = 0$
SLO No.	5.1.1.c
SLO Text	Solve simultaneous equations; c) one is linear and other is quadratic, i.e., $ax + by = m$; $cx^2 + dy^2 = n$;
Max Marks	3
Cognitive Level	A

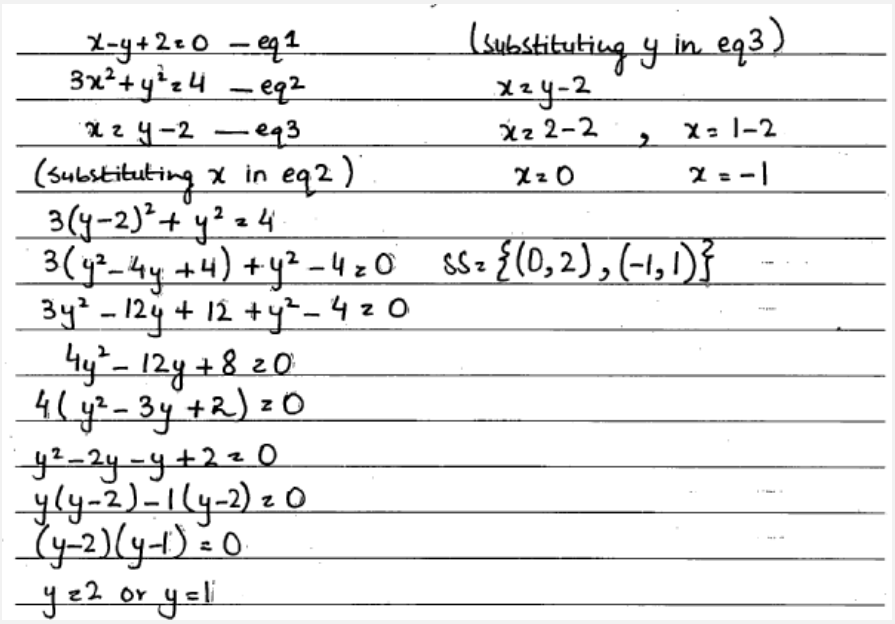
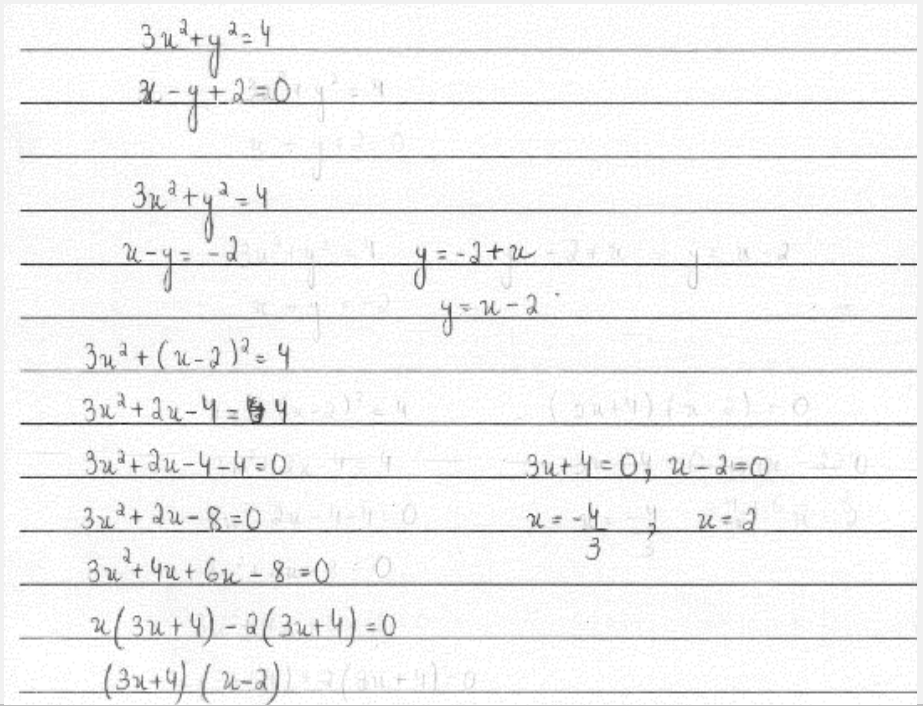
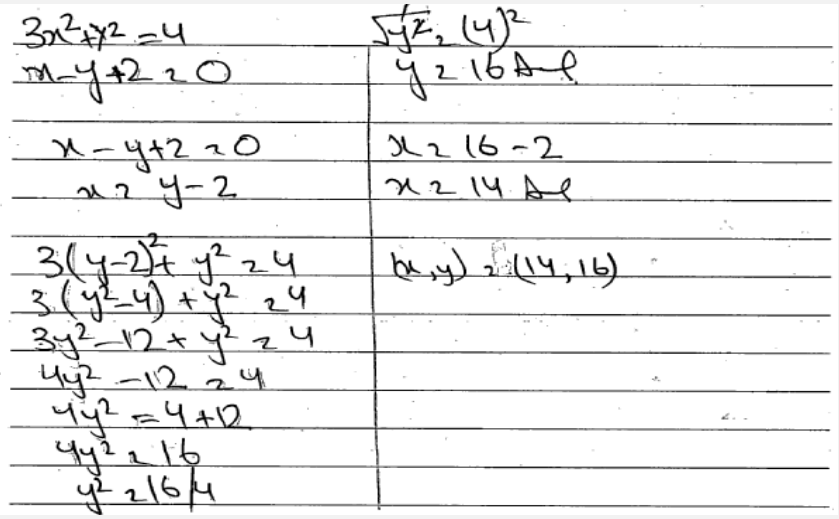
Checking Hints	1 mark for substitution for x or y 1 mark for calculating x 1 mark for calculating y
Overall Performance	Majority of candidates attempted part *a* as compared to part *b*. Candidates showed proficiency in solving simultaneous equations, especially using substitution methods. Weaker responses often involved algebraic errors and misapplication of methods.
Description of Better Responses	In better responses, candidates used the substitution method effectively, substituting $y = x + 2$ from the linear equation into the quadratic equation, resulting in $3x^2 + (x+2)^2 = 4$. They solved the resulting equation accurately to find the points of intersection.
Image of Better Response	
Description of Weaker Responses	In most of the weaker responses, candidates made errors in substitution, either substituting incorrectly or making algebraic mistakes in the resulting quadratic equation. Others struggled while solving the equation with the quadratic formula or factorisation methods, leading to incorrect solutions.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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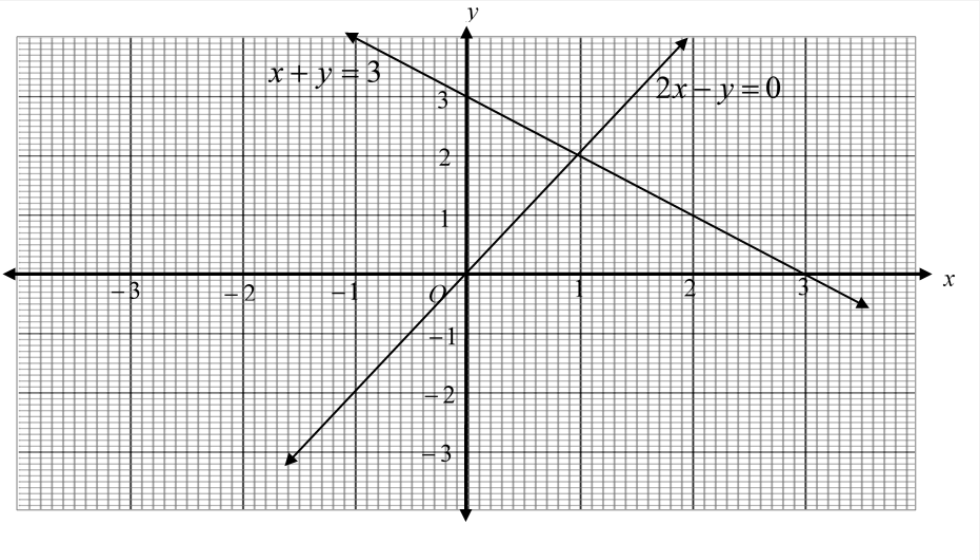
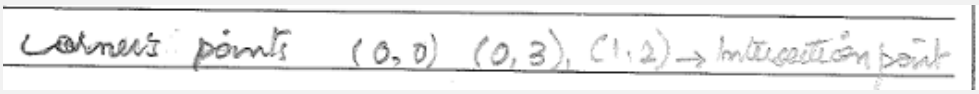
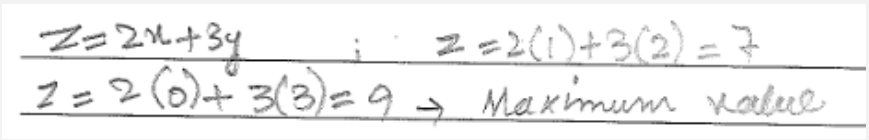
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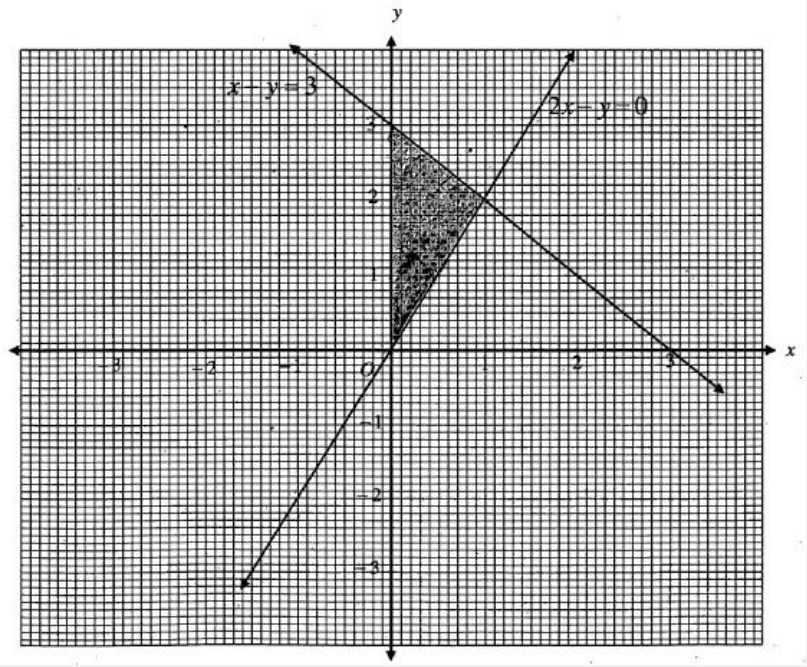
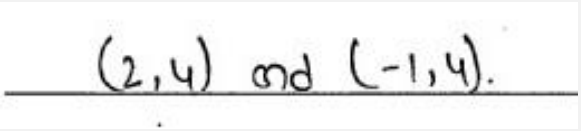
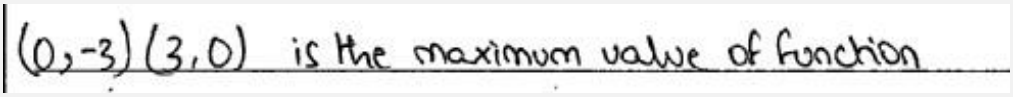
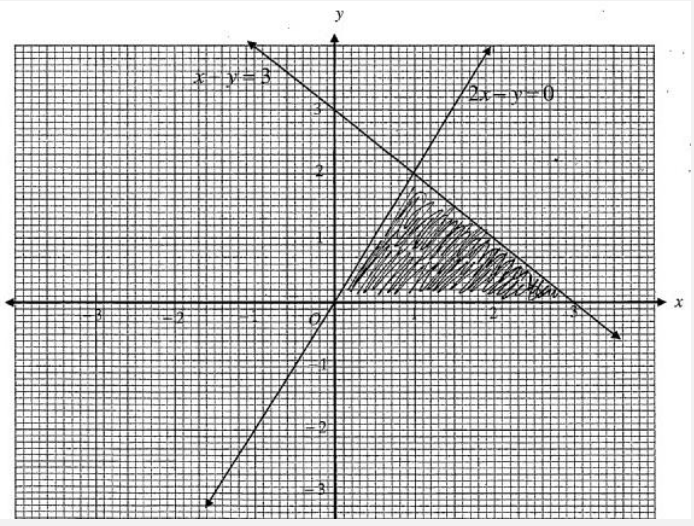
Following are some teaching strategies related to misconceptions and common errors which candidates exhibited.

- **Digital Tools:** Utilise graphing calculators or software like Desmos to visualise intersections of equations and confirm solutions.
- **Real-Life Examples:** Use problems involving economics where simultaneous equations are common, showing practical applications.
- **Peer Tutoring:** Implement peer tutoring sessions to allow students to explain their problem-solving methods to each other.

Question No. 4b

Candidates were given the choice to attempt any ONE out of the two questions: 4a and 4b.

Question Text	Maximise the objective function $Z = 2x + 3y$ subjected to the constraints $x + y \leq 3$, $2x - y \leq 0$, $x \geq 0$ and $y \geq 0$. For the given linear programing problem, i. shade the feasible region on the graph.  ii. write the corner points. iii. find the maximum value of the function.
SLO No.	5.2.3
SLO Text	Determine graphically the region bounded by three linear inequalities in two variables and the region bounded by them.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for each part
Overall Performance	A very few candidates have successfully attempted and completed all parts of the Q.4b. Others identified feasible regions or corner points but not both, indicating a partial understanding of linear programming concepts.
Description of Better Responses	In better responses, candidates correctly identified the feasible region by shading the appropriate area on the graph. They accurately identified the corner points and substituted them into the objective function to find the maximum value.
Image of Better Response	**Part (i)**  Corners points (0,0) (0,3), (1,2) → Intersection point **Part (ii)**  Z = 2x + 3y ; Z = 2(1) + 3(2) = 7 Z = 2(0) + 3(3) = 9 → Maximum value

	Part (iii) 
Description of Weaker Responses	In weaker responses, many candidates shaded the incorrect region in part (i), marking (0, 0), (3, 0), and (1, 2) instead of the correct region, which should include (0, 0), (1, 2), and (0, 3). In part (ii), many candidates were able to identify only one corner point, (3, 0), correctly however, they were unable to determine the remaining two corner points. Similarly, in part (iii), many candidates struggled to find the maximum value of the function. Instead of substituting the corner points into the given function to find the maximum value, they simply listed the corner points, and some of the candidates did not attempt this part at all.
Image of Weaker Response	**Part (i)**  **Part (ii)**  **Part (iii)** 

Suggestions for improvement (Highlight all that apply)

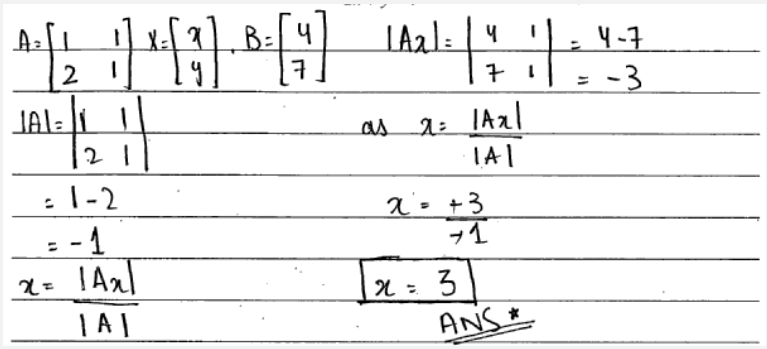
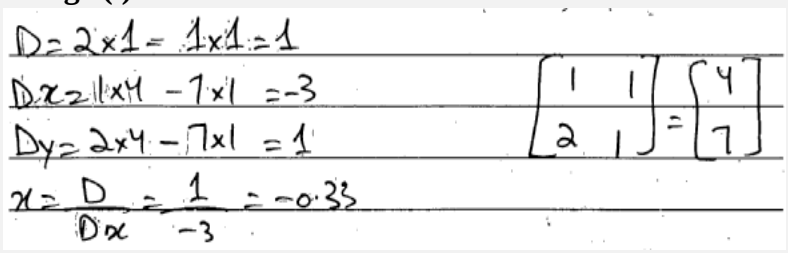
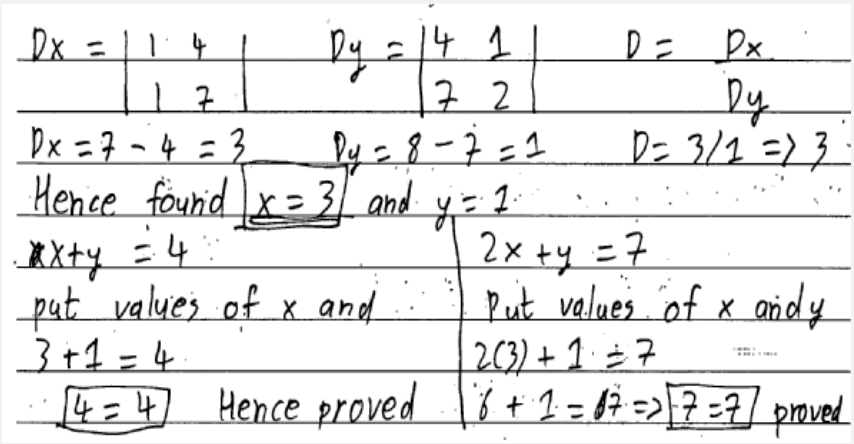
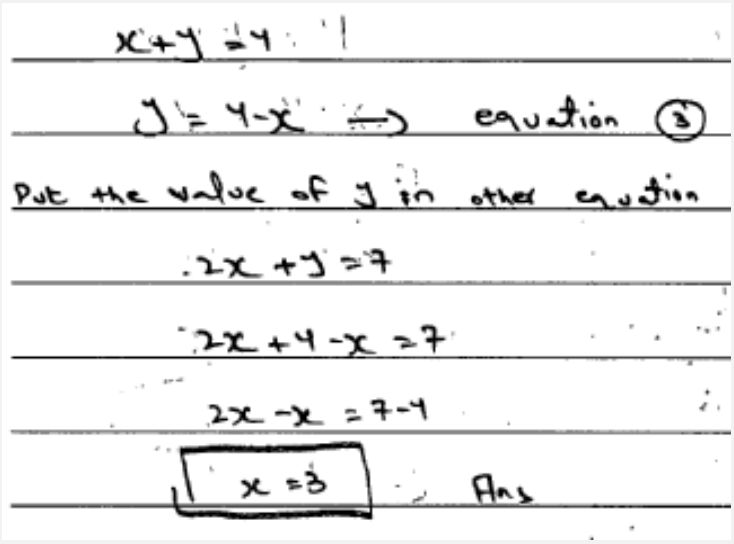
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

By implementing these teaching methodologies and strategies, teachers may create an engaging and effective learning environment that promotes a thorough understanding of this concept.

- **Digital Tools:** Use GeoGebra to practice graphing inequalities and finding feasible regions digitally.
- **Real-Life Examples:** Apply linear programming to optimise resources in business or manufacturing contexts, demonstrating practical applications.
- **Step-by-Step Guides:** Provide detailed guides on graphing inequalities and identifying feasible regions.
- **Group Problem-Solving:** Encourage group work on linear programming problems to foster collaboration and deeper understanding.

Question No. 5	
Question Text	Find the value of x by solving the given simultaneous linear equation using Cramer's rule. $x + y = 4, \quad 2x + y = 7$
SLO No.	6.4.1. b
SLO Text	Solve a matrix equation using the following methods b) Cramer's rule.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for writing equations in the form of matrices 1 mark for finding the value of $\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = -1$ 1 mark for finding the value of $x = \dfrac{\begin{vmatrix} 4 & 1 \\ 7 & 1 \end{vmatrix}}{-1} = 3$
Overall Performance	Most candidates successfully applied Cramer's rule, with few calculation errors. Weaker responses indicate a lack of understanding of the rule and its application.
Description of Better Responses	Most of candidates demonstrated a solid understanding of Cramer's rule by correctly setting up the matrices and calculating the determinants of A and A_x or D and D_x. They found the value of x accurately.

Image of Better Response	
Description of Weaker Responses	In the weaker responses, Candidates often used substitution methods incorrectly or made errors in applying Cramer's rule. Many made calculation errors with determinants, using (bc−ad) instead of (ad−bc). Others set up the matrices incorrectly, replacing the constant term with *y* while keeping *x* the same, or vice versa, for the matrices of *Dx* and *Dy*. Additionally, some candidates divided *D* by D_x instead of D_x by *D*.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

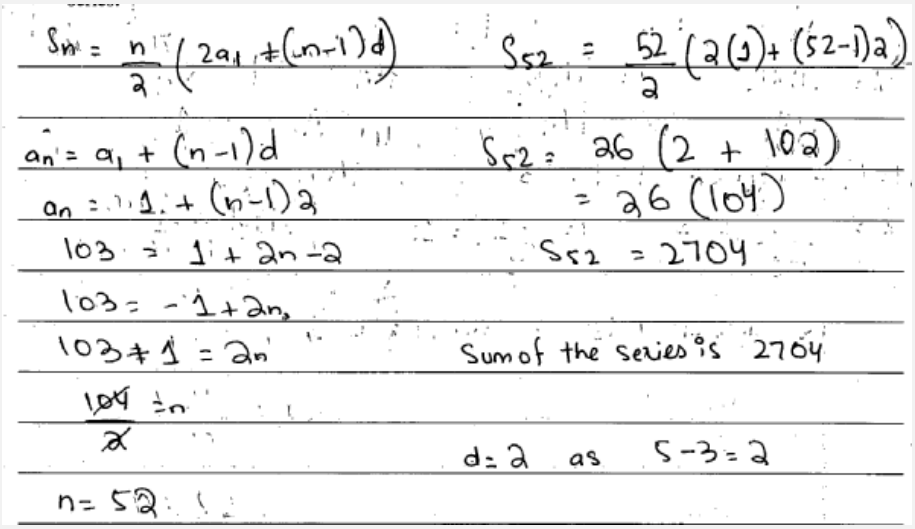
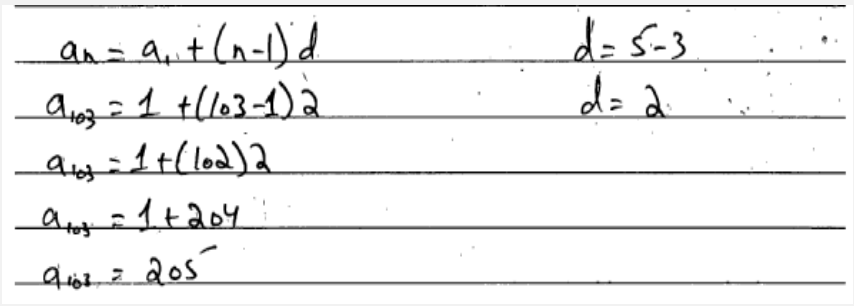
Any Additional Suggestion:

To enhance students' understanding of matrix multiplication here are a few teaching methodologies which teachers may use.

- **Digital Tools:** Use matrix calculators to practice solving systems of equations using Cramer's rule.
- **Real-Life Examples:** Show different ways through which Cramer's rule can be used to solve real-world problems like network flow or economic models.
- **Visual Aids:** Illustrate step-by-step breakdown of Cramer's rule through diagrams for visual reinforcement.

Question No. 6a
Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.

Question Text	Find the sum of the given arithmetic series. 1 + 3 + 5 + 7 + 9 + … + 103
SLO No.	8.2.2
SLO Text	Solve problems involving arithmetic series.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for the finding the value of n 1 mark for substituting the correct values in formula $S_n = \frac{n}{2}\{2a + (n-1)d\}$ 1 mark for finding the sum
Overall Performance	Many of the candidates generally performed well in this part as compared to part *b*, with a strong understanding and correct application of formulas. Weaker responses showed confusion with series formulas and calculation steps.
Description of Better Responses	In better responses, candidates demonstrated a strong grasp of series by identifying the common difference ($d = 2$), the first term ($a = 1$) and calculating the correct number of terms ($n = 52$) from the general formula $a_n = a + (n-1)d$. They applied the formula for the sum of an arithmetic series $S_n = \frac{n}{2}\{2a + (n-1)d\}$ correctly and calculated the sum accurately.

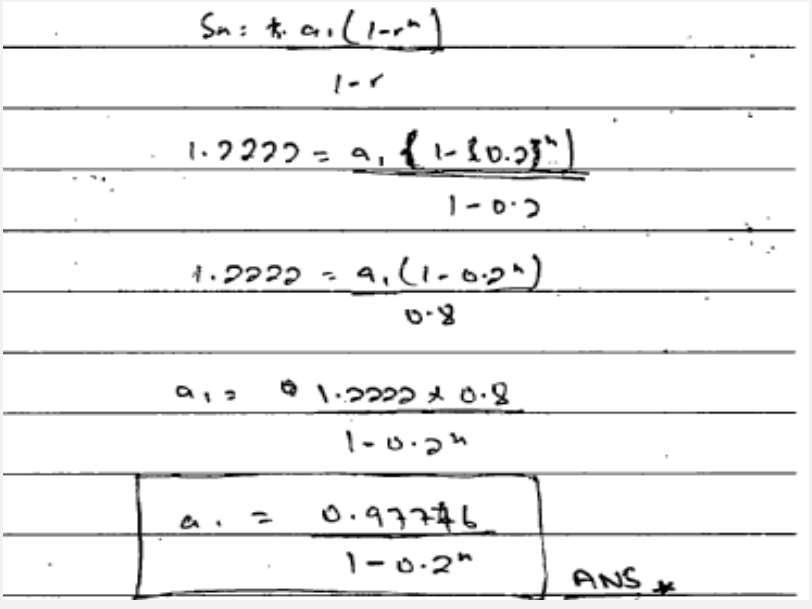
Image of Better Response	
Description of Weaker Responses	In weaker responses, candidates generally misunderstood the question. They either used the formula $a_n = a + (n-1)d$ instead of using $S_n = \frac{n}{2}\{2a + (n-1)d\}$ or failed to identify the correct common difference and number of terms. Calculation errors were also common.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:
To enhance students' understanding of matrix multiplication here are a few teaching methodologies which teachers may use.

- **Digital Tools:** Use sequence calculators or educational software to explore arithmetic series and visualise sums.
- **Real-Life Examples:** Apply arithmetic series to problems like calculating total savings over time with regular deposits.
- **Interactive Activities:** Engage students in activities where they create their own arithmetic series and calculate sums.
- **Homework Assignments:** Provide a variety of series problems to reinforce understanding and practical application of the concepts.

Question No. 6b **Candidates were given the choice to attempt any ONE out of the two questions: 6a and 6b.**	
Question Text	The sum of the finite geometric series is 1.2222 and r is equal to 0.2. Find the first term of the series.
SLO No.	8.4.2
SLO Text	Find the sum to n terms of a finite geometric series.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for the substitution in the formula $S_n = \frac{a(1-r^n)}{1-r}$ 1 mark for simplification till $0.97776 = a(1-(0.2)^n)$ 1 mark for finding the value of a as $a = \frac{0.97776}{1-(0.2)^n}$
Overall Performance	A very small number of the candidates attempted this part correctly. Their responses show a good understanding of geometric series. Weaker responses highlighted confusion with formulas and the concept of common ratio.
Description of Better Responses	In better responses, candidates applied the correct formula $S_n = \frac{a(1-r^n)}{1-r}$ and solved for the first term a in terms of n. They accurately calculated the sum, demonstrating a strong understanding of geometric series and their properties.
Image of Better Response	
Description of Weaker Responses	In most weaker responses, candidates either could not identify the correct formula or struggled to substitute the values correctly. Many substituted 1.2222 as the first term instead of S_n. Additionally, there was a frequent misunderstanding of the concept of the common ratio, along with frequent calculation errors.

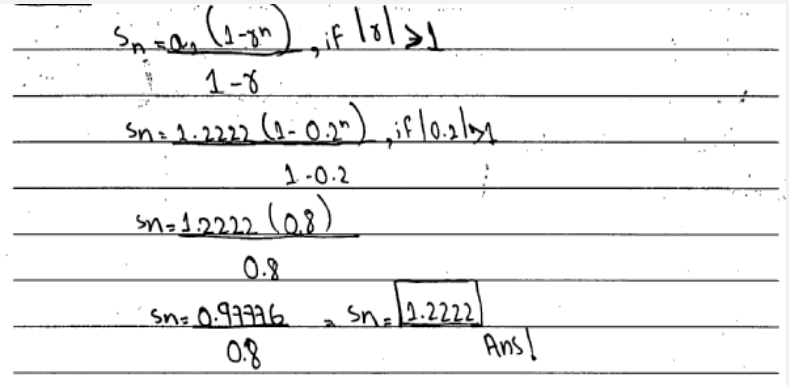
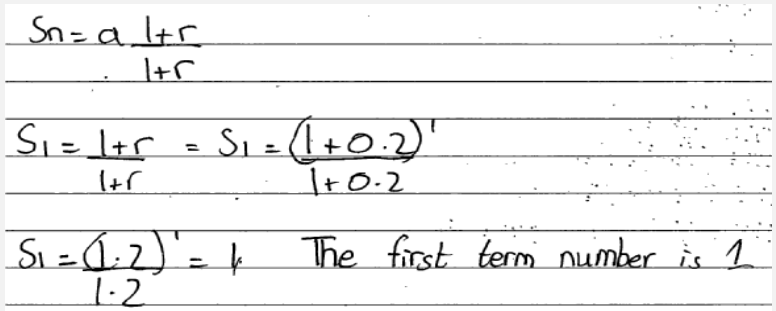
Images of Weaker Response	**Image (i)**  $S_n = \frac{a_1(1-r^n)}{1-r}$, if $\lvert r\rvert \geq 1$ $Sn = \frac{1.2222(1-0.2^n)}{1-0.2}$, if $\lvert 0.2\rvert < 1$ $Sn = \frac{1.2222(0.8)}{0.8}$ $Sn = \frac{0.97776}{0.8}$, $Sn = \boxed{1.2222}$ Ans!

Image (ii)



$Sn = a\frac{1+r}{1+r}$

$S_1 = \frac{1+r}{1+r} = S_1 = \frac{(1+0.2)^1}{1+0.2}$

$S_1 = \frac{(1.2)^1}{1.2} = 1$ The first term number is 1

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
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Any Additional Suggestion:

Following are some interactive teaching strategies for teaching sum of geometric series that teachers may use.

- **Digital Tools:** Use online tools to explore geometric series and their sums, providing visual aids for better understanding.
- **Real-Life Examples:** Apply geometric series to financial problems like calculating interest on investments with periodic additions.
- **Classroom Discussions:** Discuss different types of series and their applications in various fields to provide a broader context.
- **Practice Problems:** Provide extensive practice problems with varying degrees of difficulty to build proficiency in solving geometric series.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

The Aga Khan University Examination Board (AKU-EB) acknowledges with gratitude the invaluable contributions of all the dedicated individuals who have played a pivotal role in the development of the Business Mathematics HSSC-I E-Marking Notes.

We extend our sincere appreciation to Ms Shahneel Jawaid and Mr Muhammad Kasshaf Shaikh, Specialists in Mathematics at AKU-EB, for taking subject lead during the entire process of e-marking.

We particularly thank to Ms Nazra Naseem, Habib Girls High School, Karachi and Mr Saib Ahmed Khan, Govt. Degree Science and Commerce College Orangi Township, Karachi on evaluating each question's performances, delineating strengths and weaknesses in candidates' responses, and highlighting instructional approaches along with recommendations for better performance.

Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.
These contributors include:

- Mahrukh Amin Jiwa, Specialist, Curriculum Development
- Zain Muluk, Manager, Examination Development
- Raabia Hirani, Manager, Curriculum Development
- Dr Naveed Yousuf, CEO, AKU-EB